Want to show:

$$\left(\forall\ \Sigma,\ A\ \Sigma\text{-alg},\ W\text{-eq ax}\right.$$

$$A \vDash ax \Rightarrow Con_S A \vDash ax$$

$$\Rightarrow S \text{ is } W\text{-compatible.}$$

Assume. Take X

Define: Σ'; $(\rho : \Gamma \to \Delta)$

$$\ulcorner \rho \urcorner : \blacksquare \to 1$$

$$A := (n : \mathbb{N}) \times X^n$$

$$[\![\rho]\!] : (n_{y} : \mathbb{N}, (a_i^{y})_{i<n})_{y \in \blacksquare} \longmapsto$$

$(n:\mathbb{N}) \times \mathbb{N} \cong \mathbb{N}$

$$\left(\langle (y \in \Delta,\ n_{y[\rho]} \rangle,\ \mapsto a_i^{y[\rho]}\right) \qquad (y,\ i < n_{y[\rho]})$$

$$\ulcorner \rho \circ \sigma \urcorner = \ulcorner \rho \urcorner [\sigma]$$